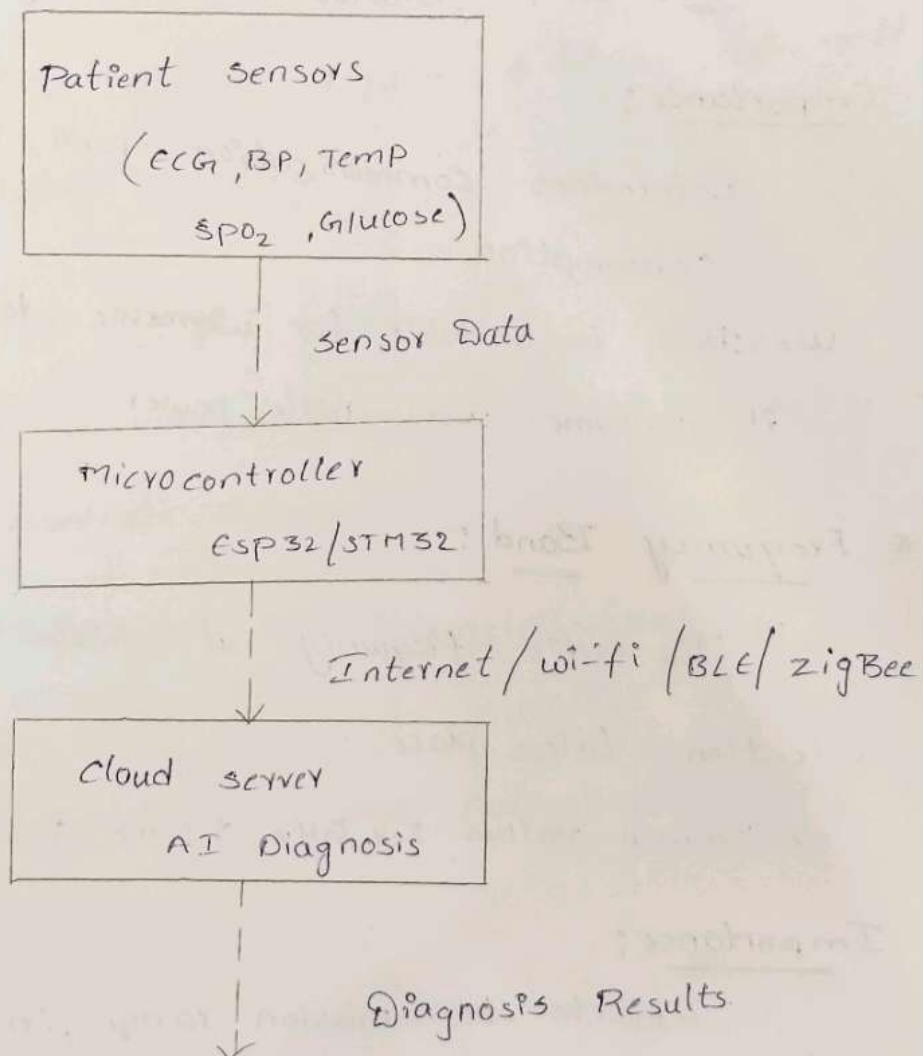


1) Aim:

Design a wireless communication workflow for an AI-assisted healthcare diagnosis system that collects patient data, transmits it wirelessly, processes it using AI, and sends diagnostic results back to healthcare professionals while meeting communication requirements.

Work flow



Doctor Dashboard
mobile APP/web

Parameters:

* Communication Technology:

⇒ The wireless protocol used to transmit patient data between sensors, gateways, and cloud servers.

Ex: wi-fi, Bluetooth Low energy (BLE), LoRa, 5G.

Importance:

Determines communication speed, range and power consumption.

uses: BLE is suitable for wearable devices because it consumes very little power.

* Frequency Band:

⇒ The radio frequency at which wireless communication takes place.

⇒ Common values 2.4 GHz, 5 GHz, Sub-6 GHz (5G)

Importance:

Affects transmission range, interference and data rate.

Ex: Wi-fi commonly operates at 2.4 GHz and 5 GHz.

* Data Rate:

⇒ The amount of data transmitted per second.

unit: kbps (kilobit/s) (or)

mbps (mega bits/s)

Importance:

Higher data rates allow faster transmission of medical information.

Ex: ECG signals require a higher data rate than temperature readings.

* Transmission Range:

⇒ The maximum distance over which wireless devices can communicate reliably.

unit: meters (m) (or) kilometers (km)

Importance:

Determines whether patient monitoring is local or remote.

Ex:

BLE has a range of about 10-30m, while wi-fi can reach around 100m indoors.

* Latency:

⇒ The time taken for data to travel from Patient device to the doctor or cloud server.

unit: milliseconds (ms).

Importance: critical for emergency healthcare applications.

Ex: Heart attack alerts require latency below 50ms.

* Packet Delivery Ratio (PDR):

⇒ The Percentage of transmitted data packets that are successfully received.

$$PDR = \frac{\text{Packets Received}}{\text{Packets sent}} \times 100\%$$

Importance: Higher PDR means more reliable communication.

Ex:

if 100 packets are sent and 98 are received,
the PDR is 98%.

* Bandwidth:

→ The maximum amount of data that can be transmitted through the communication channel.

unit: MHz (or) mbps

Importance:

Higher bandwidth supports multiple medical devices simultaneously.

Ex: wi-fi provides greater bandwidth than Bluetooth.

* Throughput:

The actual amount of useful data successfully transmitted over the network.

unit: mbps

Importance:

Indicates real communication performance after accounting for delays and retransmissions.

Ex:

A network with a 100 mbps data rate may achieve only 80 mbps throughput.

* Bit Error Rate (BER):

⇒ The ratio of incorrectly received bits to the total number of transmitted bits.

$$\text{BER} = \frac{\text{Number of Error Bits}}{\text{Total number of Bits sent.}}$$

Importance:

Lower BER indicates better communication quality.

Ex:

A BER of 10^{-6} means only one bit error occurs in one million transmitted bits.

Availability:

⇒ The percentage of time the communication system is operational and accessible.

$$\text{Availability} = \frac{\text{system uptime}}{\text{Total time}} \times 100\%$$

Importance:

High availability ensures uninterrupted patient monitoring.

Ex: A healthcare network with 99.99% availability experiences very little downtime.

* Reliability:

⇒ The ability of the communication system to consistently deliver accurate data without interruption

Importance:

Essential for continuous patient monitoring and emergency alerts.

Ex: A reliable system ensures ECG data reaches the doctor without loss.

Working steps:

Step 1: Data collection

⇒ Analog signals are converted into digital data.

Step 2 : Data processing

⇒ microcontroller filters noise and prepares sensor packets.

Step 3 : wireless Transmission

⇒ sensor data is transmitted using wi-fi, BLE, ZigBee, 5G.

Step 4 : Cloud communication

⇒ gateway forwards the data through the internet.

Step 5 : AI Diagnosis

⇒ AI model analyzes sensor values

⇒ predicts disease risk.

Step 6 : Result

⇒ Diagnosis report is transmitted back wirelessly.

⇒ Doctor receives alerts through a mobile application or web dashboard.

Advantages :

- Real-time patient monitoring.
- Fast AI-assisted diagnosis.
- Reduced hospital visits.
- Continuous health monitoring.
- Remote wireless communication.
- Early disease detection.

Challenges :

- wireless interference.
- Battery limitations.
- Network congestion.
- Data privacy concerns
- packet loss
- AI prediction accuracy
- system maintenance.
- Internet dependency.

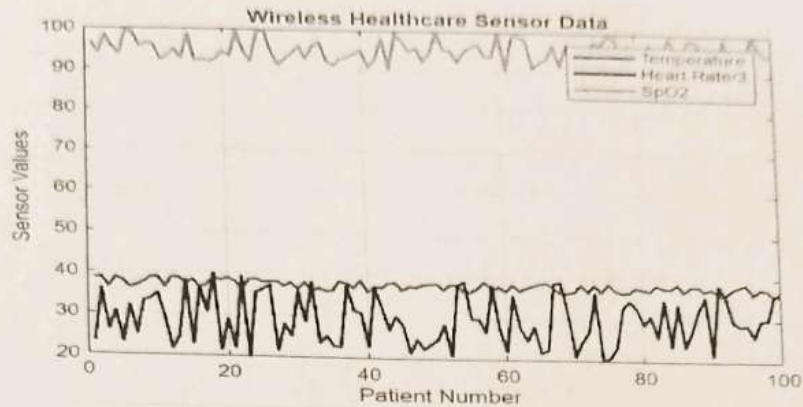
Outcome:

The designed workflow enables reliable wireless transmission of patient health data to an AI-based diagnosis system with low latency, high reliability, secure communication, minimal power consumption, and accurate real-time healthcare monitoring.

Code:

```
clc;  
clear;  
close all;  
N=100;  
Temperature = 36 + rand(1,N)*3;  
heart rate = 60 + rand(1,N)*60;  
SpO2 = 90 + rand(1,N)*10;  
delay = rand(1,N)*5;  
Diagnosis = zeros(1,N);  
for i=1:N  
    if (Temperature(i) > 37.5 || heart rate(i) > 100 || SpO2(i) < 95)  
        else  
            diagnosis(i) = 1;  
        end  
    end  
end
```

Patient	Temp	HR	SpO2	Delay(ms)	Diagnosis
1	38.44	69.7	96.4	0.30	Abnormal
2	38.72	107.7	93.8	3.41	Abnormal
3	36.38	78.7	98.1	0.23	Normal
4	38.74	91.7	95.3	0.36	Abnormal
5	37.90	69.9	93.5	2.61	Abnormal
6	36.29	96.1	99.4	0.48	Normal
7	36.84	75.8	98.8	4.09	Normal
8	37.64	99.2	95.5	4.09	Abnormal
9	38.87	101.4	96.2	3.61	Abnormal
10	38.89	104.9	95.9	0.75	Abnormal



```
clc;
clear;
close all;
```

```
% Number of patients
N = 100;
```

```
% Generate Sensor Data
```

```
Temperature = 36 + rand(1,N)*3; % 36-39 °C
HeartRate = 60 + rand(1,N)*60; % 60-120 bpm
SpO2 = 90 + rand(1,N)*10; % 90-100 %
```

```
% Wireless Transmission Delay (5G)
```

```
delay = rand(1,N)*5; % 0-5 ms
```

```
% AI Diagnosis
```

```
Diagnosis = zeros(1,N);
```

```
for i=1:N
```

```
if (Temperature(i)>37.5 || HeartRate(i)>100 || SpO2(i)<95)
```

```
Diagnosis(i)=1; % Patient Needs Attention
```

```
else
```

```
Diagnosis(i)=0; % Normal
```

```
end
```

```
end
```

